MathsJam Shout

April 2026

Astoria MathJam, OR USA

Puzzle
Number Search

Combining a wordsearch with a crossword puzzle and using numbers instead of letters, you get Number Search, with homemade clues made from randomly generated grids of numbers.

More puzzles: astoriabob.net/numbersearch

2	8	8	6	6	6	6
1	9	9	1	3	6	4
5	8	2	2	1	7	2
4	3	2	2	5	4	8
8	6	8	1	0	6	8
4	2	0	6	0	7	9
5	7	7	5	9	8	7

Clues

Call up Copper Rock Golf Course in Hurricane UT

1100010000011000111000012

MMCCXCII

Backwards and forwards multiple of 19 (4 digits)

Area under the curve $\frac{\ln x}{x}$ from x = 1 to $e^{\sqrt{158}}$

First part of who do we appreciate without 8

Looks like double fortune or happiness in Chinese (囍)

Game
Take Sides

The goal of the game is to design an equation using only the numbers on all 5 dice (once each), using arithmetic operators (including exponents) and concatenation.

Here are two solutions (there are more) for this example roll:

$3^{(3-2)} + 2 = 5$

$2 \times 2 = 5 - \frac{3}{3}$

Try it yourself! For fun, try different numbers of dice and see if it's easier or harder.

More info: mathsj.am/takesides

Puzzle
Yohaku

Using either addition or multiplication (indicated with the lower right symbol), find values of a–i that give entries which add to the numbers at the end of the row or column.

$\frac{1}{a}$	$\frac{1}{b}$	$\frac{1}{c}$	$\frac{11}{20}$
$\frac{1}{d}$	$\frac{1}{e}$	$\frac{1}{f}$	$\frac{17}{24}$
$\frac{1}{g}$	$\frac{1}{h}$	$\frac{1}{i}$	$\frac{131}{231}$
$\frac{11}{24}$	$\frac{31}{30}$	$\frac{181}{660}$	+

The solution for this grid is hidden at the bottom of the page.

Can you come up with your own puzzles to share, by reverse engineering?

More info: yohaku.ca

Think
Unit Fractions

Can you arrange the numerals 1 to 9 in a single fraction that equals exactly 1/3? Can you solve this for all the unit fractions with a denominator from 2 to 19? Which ones can't be done? Which ones have unique solutions?

Solve
Alphametics

For each sum, assign the digits 0–9 to the letters so the phrase is correct arithmetically.

WHERE + IS + THE + REAL = TRUTH

I + STILL + WANT + TO + STAY = SILLY

A + THIRD + TIMES + A + THIRD = NINTH

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com**

Maths Jam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly Maths Jam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest Maths Jam, visit the website at **mathsjam.com.**

a = 4, b = 5, c = 10, d = 8, e = 2, f = 12, g = 7, h = 3, i = 11